

Exercise Sheet - 1

STA201: Elements of Statistics & Probability

Question 1

The Connecticut Agricultural Experiment Station conducted a study of the calorie content of different types of beer. The calorie content (calories per 100 ml) for 26 brands of light beer are:

29, 28, 33, 31, 30, 33, 30, 28, 27, 41, 39, 31, 29, 23, 32, 31, 32, 19, 40, 22, 34, 31, 42, 35, 29, 43

- (i) Construct a stem-and-leaf display using stems 1, 2, 3, and 4.
- (ii) If we organized this data in a frequency distribution, how many classes would you recommend? What class interval would you suggest?
- (iii) Find a suitable average. Replace 42 by 92, then again find the average and comment.

Question 2

Three professors at BRAC University compared two different approaches to teaching courses in the School of Data and Sciences. At the time of the study, there were 2,100 students in the School of Data and Sciences, and 96 students were involved in the study. Demographic data collected on these 96 students included class (freshman, sophomore, junior, senior), age, gender, and major.

- a) What is the population of interest and the sample that was collected?
- b) For each of the four demographic variables mentioned, indicate the measurement scale.

Question 3

You are working for the transport manager of a call center which hires cars for the staff. You are interested in the weekly distances covered by these cars. Kilometers recorded for a sample of hired cars during a given week yielded the following data:

Kilometers Covered	No. of Cars	Kilometers Covered	No. of Cars
100–110	4	150–160	8
110–120	0	160–170	5
120–130	3	170–180	0
130–140	7	180–190	2
140–150	11		

- What type of grouping has been done here?
- Find the average distance using a suitable method.
- Draw a histogram and frequency polygon.
- Draw an ogive curve for the above data.

Question 4

The following sample data show the mileage of cars:

Mileage (km/litre)	No. of Cars
0–6	7
6–12	13
12–18	6
18–24	5
24–30	4

- Find a suitable mean of the above data and comment.

Question 5

The following frequency distribution represents the prices of different items at a departmental store:

Price (Tk.)	Number of Items
120–130	5
130–140	6
140–150	14
150–160	10
160–170	5

- Draw a frequency polygon and histogram for the price of items.
- Find the suitable mean for the above data.

Question 6

The growth rates of production of a manufacturing factory compared to last year are given below:

Month (2014)	Growth Rate (%)
January	5
February	7.5
March	6
April	4
May	3.2
June	8

Represent the growth rate by a suitable graph and comment. Also find the suitable average of growth rate.

Question 7

Thirty adults were asked which of the following conveniences they would find most difficult to do without: television (T), refrigerator (R), air conditioning (A), public transportation (P), or microwave (M). Their responses are listed below:

R A R P P T R M P A A A R T P P T R A A R P A T R P R A A R

- Prepare a frequency distribution table.
- Calculate the relative frequencies and percentages for all categories.
- Draw a bar graph for the frequency distribution.

Question 8

A Crude Oil Company has a number of outlets in the Dhaka metropolitan area. The numbers of oil changes at the Green Street outlet in the past 25 days are:

65, 98, 55, 62, 79, 59, 70, 62, 66, 80, 94, 79, 51, 63, 66, 55, 67, 67, 87, 78, 87, 84, 78, 83, 69

- Construct a stem-and-leaf display using stems 5, 6, 7, 8 etc.
- If we organized this data in a frequency distribution, how many classes would you recommend? What class interval would you suggest?

Question 9

In a survey of 100 people who had recently purchased motorcycles, data on the following variables was recorded: Gender of purchaser, Brand of motorcycle purchased, Number of previous motorcycles owned by purchaser, Telephone area code of purchaser, Weight of motorcycle as equipped at purchase.

- a) Which of these variables are categorical? Which are discrete numerical?
- b) Determine the level of measurement for each of the variables.

Question 10

The LG Manufacturing Company makes electric wiring, which it sells to contractors in the construction industry. Approximately 900 electric contractors purchase wire from LG annually. LG's director of marketing wants to determine the electric contractor's satisfaction with LG wire. He developed a questionnaire that yields a satisfaction score between 10 and 50 for participant responses. A random sample of 35 of the 900 contractors is asked to complete a satisfaction survey. The satisfaction scores for the 35 participants are averaged to produce a mean satisfaction score.

- a) What is the population and sample for this study?
- b) What would be a parameter for this study and what is the statistic?

Question 11

Lives of two models of Refrigerators in recent survey are given below:

Life in year	0–2	2–4	4–6	6–8	8–10	Total
Singer	5	13	20	X	5	50
Konka	K	7	12	20	9	50

- i) Find the value of X & K. Which model has more average life? Give your decision with finding suitable positional average.
- ii) A person wants to buy a refrigerator. Which one should they prefer and why?

Question 12

The Citizens Banking Company is studying the number of times the ATM located in a Supermarket at the foot of Market Street is used per day. Following are the numbers of times the machine was used over each of the last 30 days:

83, 64, 63, 80, 95, 36, 84, 84, 78, 76, 73, 61, 84, 68, 59, 54, 52, 84, 75, 65, 95, 59, 90, 47, 70, 52, 87, 61, 77, 60

- i) Construct a stem-and-leaf display using stems 3, 4, 5 etc.
- ii) If we organized this data in a frequency distribution, how many classes would you recommend? What class interval would you suggest?

Question 13

Refer to the Baseball 2022 data, which reports information on the 30 Major League Baseball teams for the 2022 season. Consider the following variables: number of wins, team salary, season attendance, whether the team played its home games on a grass field or an artificial surface, and the number of home runs hit.

- a) Which of these variables are quantitative and which are qualitative? Which of these variables are discrete numerical?
- b) Determine the level of measurement for each of the variables.
- c) Which type of graphical display would be an appropriate choice for summarizing the given data? Give a list of charts for each of the given variables.

Question 14

The distribution of production of electricity (in 000 MW) in different days is shown below:

Class Interval of Production	Number of Days, f
2.0–2.4	5
2.4–2.8	18
2.8–3.2	28
3.2–3.6	42
3.6–4.0	7

- a) Calculate mean production of electricity.

Question 15

The Walton Manufacturing Company makes electric wiring, which it sells to the construction industry. Approximately 900 electric contractors purchase wire from Walton annually. Walton's director of marketing wants to determine the electric contractor's satisfaction with Walton wire. He developed a questionnaire that yields a satisfaction score between 10 and 50 for participant responses. A random sample of 35 of the 900 were asked to complete a satisfaction survey. The satisfaction scores for the 35 participants are averaged to produce a mean satisfaction score.

- i) What is the population and sample for this study?
- ii) What would be a parameter for this study and what is the statistic?

Question 16

The following data show the method of payment by 16 customers in a supermarket checkout line. Here, C refers to cash, CK to check, CC to credit card, D to debit card, and O stands for other:

C, CK, CK, C, CC, D, O, C, CK, CC, D, CC, C, CK, CK, CC

- i) Construct a frequency distribution table.
- ii) Draw a bar chart for the percentage distribution.

Question 17

A nurse records the respiratory rates (in beats per minute) of female patients and male patients in a hospital:

Female: 95, 89, 106, 92, 114, 96, 107, 89, 115, 91

Male: 72, 83, 67, 75, 80, 87, 68, 76, 91, 72

- i) Calculate the suitable average of respiratory rates for female and male patient groups. For which group is the respiratory rate larger?